Mathematical overview: facts, assumptions and limits

A journey is a finite ordered walk on a fixed graph, and shortening deletes only adjacent opposite traversals of the same road. Such reduction is unique. Every tree walk from one vertex to another reduces to their unique simple path; closed tree walks reduce to empty. New roads change the graph, so reduced closed walks may persist. A vertex of degree one cannot appear in a nonempty reduced closed journey, because visiting it creates an immediate reversal.

On the four-arm star in P1, each of the four leaves must acquire another incident road to occur in a reduced closed tour. Each new road touches only two leaves, so at least two additions are necessary. A perfect matching of the four leaves attains the bound. The resulting two triangles sharing H admit a six-step reduced tour through all four leaves, and six is optimal when there are only two added roads. Removing either addition leaves two degree-one outer stops and prevents any reduced closed tour visiting all four, although the other triangle still supports a surviving tour.

Return-walk counts concern unreduced histories. On a three-arm unit star, six-step closed walks are three successive outward/return excursions, hence $3^3=27$. On a star with each arm extended by one edge, six-step closed walks include nested excursions and number 48. Every one still reduces to empty. Equal reduced journeys therefore do not imply equal counts of unreduced histories.

At a two-ring junction, whole loop words a and b freely concatenate. Capital A/B denote the corresponding reverse loop, so adjacent aA/Aa/bB/Bb cancel. Distinct loops a,b do not commute, but powers of a common reduced word do. The supplied finite card collection has exactly three commuting pairs. This finite result does not require the full free-group centralizer theorem; that theorem is adult context, not a K–5 prerequisite.

Grades 2–3 have physical route design and finite counting entries; Grades 4–5 can prove optimality/completeness and explore order trading. Prerequisites are replaying an ordered walk, recognizing immediate opposite road steps, counting to six, and maintaining an unchanged map. Macro words require reliable bridge from each symbol to a whole loop. No independent reading or abstract group notation is needed if an adult reads the task, but the child must own the route choices. No forced K–1 page is added to these investigations; the base pawn activities remain the concrete entry if this readiness gate is not met.

Materials, launch and pacing

Per pair: two large tree mats (the P1 four-arm star and P2 extended star), a reusable P2 unit-star mat or the other side of a sheet, the base-style two-triangle figure-eight mat relabeled H/L/M/R/S, two pawns, 24 directed road-step cards with room for origin/destination labels, seven whole-journey cards carrying a, aa, b, ab, abab, abA, abbA, six removable road strips, pencil/eraser and a tray. A blank card may be relabeled for another road; cards need not require a full preprinted deck of all walks. The seven whole-journey cards are selectors: use them to choose and order trips, then expand one concatenation at a time into individual road-step cards. The longest pair contains eight whole loops, hence 24 road steps, so one expanded row fits the supplied deck. Save that row's reduced route or word on paper (or in a photograph), reuse the 24 step cards for the other order, and compare its reduction with the saved record. Two simultaneous expanded rows are not part of this kit. In P2 keep one recoverable ordered walk record, then count its six steps. Use a partner as recorder/referee and rotate. Six full pair kits for the eleven-child K–1 / 2–3 / 4–5 roster require 24 mats (four kinds each), 12 pawns, 144 step cards, 42 word cards, 36 road strips and six trays. Actual use can be restricted to three active older-table kits while K–1 revisits base concrete pawn work. Three anchored adults suffice when mat/card preparation is done ahead.

Student P1/P2 pawn circles are 58 pt =20.5 mm diameter. P3's map is compact because children use separate mats and movable word cards; it is a code key, not a working pawn board. Draw separate road mats with at least 70 mm between vertex centers. A new A–C road may curve around H: touching its printed H dot would introduce a different road junction, so keep it clear. Crossings without dots are not intersections.

Launch: let children move pawns and handle route cards. On a non-task two-road chain, walk out, back, then onward; remove the adjacent out/back pair and replay remaining cards. Confirm that

removing a detour keeps the rest in order. Before P3 introduce a as H-L-M-H and physically run a backward as A=H-M-L-H. The printed aAb example has input, removal and output b; demonstrate the three individual inverse road cancellations within aA so whole-loop cancellation has a reason. Keep the same start H throughout. Carry that convention into composite words such as abA: cancelling a touching inverse symbol inside it does not remove its whole selector card. Remove only the corresponding individual road steps, or write the shortened word in order.

P1 can fill 20–35 minutes with designs and optimality. P2 can fill 20–40 minutes, especially if pairs share branch cases rather than each copying 48 routes. P3 warrants its own 20–40-minute return visit. Pool a checkable collection at each fixed table; no table mixing or full individual transcription is required. All timing and pawn/card handling remain unpiloted. Exact material fit and transfer have **not** been physically pretested. An adult should rehearse the rule/card bridge, and if children cannot replay ordered cards, remain with the concrete base launch.

Solutions, hints and extensions

P1 / student p.1. Two new roads are optimal. Exactly three unordered optimal additions exist: AB+CD, AC+BD, or AD+BC. A six-step witness for AB+CD is H-A-B-H-C-D-H. The other pairings have analogous tours. There are no immediate reversals. At least two roads are required because every outer stop needs a second incidence; with only two roads they must pair all four leaves. In that graph each excursion from H that survives is at least a triangle of 3 steps. Visiting both triangles requires at least 6 steps; the witness attains it. Closing either new road turns its two leaves back into degree-one stops. A nonempty reduced trip may survive around the other triangle but cannot visit all four outer stops. The phrase “keep every step after shortening” excludes adding disposable leaf excursions to a surviving triangle. Hint: try entering and leaving a leaf without using the same road backward. Extension: allow three added roads, or require a reduced tour that visits H only at start/finish; a five-step cycle through H and all four leaves uses three additions. Verify actual road identities rather than count apparent pencil crossings.

P2 / student p.2. Left unit star:27 six-step returns. Right extended star:48. A complete explanation groups walks by lengths between successive returns to H. In the unit star all returns consist of three two-step excursions:3 choices each. In the extended star: (2,2,2) gives 27; (2,4) and (4,2) give 3×3 each, total 18; one six-step excursion with no intermediate H return is H-A-D-A-D-A-H or its B/E and C/F versions, total 3. Sum 27+18+3=48. This grouping is adult scaffolding to offer only after children create examples. The raw words can be pooled by first arm or first return; a photograph/ordered route record is enough, without simultaneous tallies. Every trip reduces to empty because both maps are trees. Hint: ask whether there are walks that visit D/E/F before returning H. Extension: four-step counts are 9 and 12; use them to check a conjecture about growth, or add just one outer road and ask how many returns use it. Do not infer the six-step result solely from a small sample.

P3 / student p.3. Treat word cards as seven specific nonempty whole trips. Exactly these unordered pairs commute after reduction:

- a with aa: both orders reduce to aaa.
- ab with abab: both orders reduce to ababab.
- abA with abbA: both orders reduce to abbbA. Here abbA is the square of abA after the middle Aa cancellation.

All other 18 pairs fail. The given aAb demonstration reduces to b. A complete finite solution may show a table of 21 tested pairs with reduced outputs; the verifier checks both macro reduction and expansion to the actual six individual ring roads. For a deeper explanation, recognize powers of a common journey: the third pair is a whole left excursion, one/two right loops, and the return along the left loop in reverse. Shared net loop counts do not suffice to determine commutation. Hint: save one reduced route/word, reuse the individual step cards for the other concatenation, then compare. Keep the word-selector cards intact; never delete a whole composite card to cancel one symbol within it. Replay a disputed reduction on the unchanged mat. Extension: create a new pair using both rings that commutes but is not literally identical, and another with the same net loop counts that does not. The full theorem that commuting nontrivial free-group elements are powers of a common word may be

supplied as adult context, never required as the elementary entry.

Distinction and verification

Base Week 39 proves tree reduction, compares ring outcomes and one commutator. P1 makes road additions/deletions an optimization design; P2 counts unreduced histories on contrasting trees instead of re-proving empty reduction; P3 classifies a finite commutation collection with shared-word examples and cancellation. These are original constructions using the base reduction rule and Week 1 encore's revisit structure; no downloaded puzzle is copied.

student/verify.py exhausts all 64 additions among the four outer vertices, finds shortest reduced closed tours with vertex/previous-edge/visited-state search, and checks every optimal deletion. It enumerates all six-step walks and checks empty reduction. It reduces all 21 word pairs and independently expands them into directed road steps to confirm every equality. Portable student/build.sh OUTPUT_DIR requires Python 3 and ReportLab, uses no repository imports, and leaves output selection explicit. Record use as W39-B1/B2/B3 with actual subquestions attempted; all material is unpiloted and physical pretests unperformed.